AP PRECALCULUS · TOPIC 1.9

# Rational Functions and Vertical Asymptotes

Section 1 — Multiple Choice
Circle the letter of the correct answer **SOLVED**

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

**1** Which of the following gives all vertical asymptotes of $r(x) = \dfrac{x+4}{x^2-x-12}$?

**A.** $x = -3$ and $x = 4$
**B.** $x = 3$ and $x = -4$
**C.** $x = -4$
**D.** $x = -4$, $x = -3$ and $x = 4$

**A.** $r(x) = \dfrac{x+4}{(x-4)(x+3)}$; the denominator is $0$ at $x = 4$ and $x = -3$ and the numerator is not $0$ at either. B: sign error, reading each factor $x - a$ as $x = -a$. C: set the numerator equal to $0$ (that gives the zero, not an asymptote). D: also counted the zero of the numerator as an asymptote.

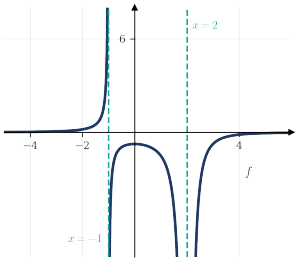
**2** Which of the following gives all vertical asymptotes of $r(x) = \dfrac{x^2-9}{x^2-x-6}$?

**A.** $x = -2$ and $x = 3$
**B.** $x = -2$ only
**C.** $x = 3$ only
**D.** $x = -3$ and $x = 2$

**B.** $r(x) = \dfrac{(x-3)(x+3)}{(x-3)(x+2)}$; the factor $x - 3$ cancels (a hole at $x = 3$) and $x + 2$ remains. A: treated every zero of the denominator as an asymptote. C: swapped the roles of the two denominator factors. D: sign error, reading $x - a$ as $x = -a$.

**3** The graph of the rational function $f$ is shown, with vertical asymptotes at $x = -1$ and $x = 2$. Which statement is true?

**A.** $\lim\limits_{x\to -1^+} f(x) = \infty$ and $\lim\limits_{x\to 2^+} f(x) = -\infty$
**B.** $\lim\limits_{x\to -1^+} f(x) = -\infty$ and $\lim\limits_{x\to 2^+} f(x) = \infty$
**C.** $\lim\limits_{x\to -1^+} f(x) = -\infty$ and $\lim\limits_{x\to 2^+} f(x) = -\infty$
**D.** $\lim\limits_{x\to -1^+} f(x) = 0$ and $\lim\limits_{x\to 2^+} f(x) = 0$




**C.** Right of $x = -1$ the branch falls without bound; right of $x = 2$ the branch also falls without bound. A: used the branch on the left of $x = -1$. B: assumed the two sides of $x = 2$ differ, as at $x = -1$. D: read the end behavior (the $x$-axis) as the behavior at the asymptotes.

**4** What is the value of $\lim\limits_{x\to 3^-} \dfrac{x+1}{x-3}$?

**A.** $\infty$
**B.** $0$
**C.** $4$
**D.** $-\infty$

**D.** The numerator tends to $4$ and $x - 3$ is a tiny negative number, so the quotient decreases without bound. A: ignored the sign of $x - 3$ on the left. B: thought a tiny denominator gives a tiny value. C: substituted $x = 3$ into the numerator only.

**5** Let $g(x) = \dfrac{5}{(x+2)^2(x-1)}$. Which statement is true?

**A.** $\lim\limits_{x\to -2^-} g(x) = -\infty$ and $\lim\limits_{x\to -2^+} g(x) = \infty$
**B.** $\lim\limits_{x\to -2^-} g(x) = -\infty$ and $\lim\limits_{x\to -2^+} g(x) = -\infty$
**C.** $\lim\limits_{x\to -2^-} g(x) = \infty$ and $\lim\limits_{x\to -2^+} g(x) = \infty$
**D.** $\lim\limits_{x\to -2^-} g(x) = \infty$ and $\lim\limits_{x\to -2^+} g(x) = -\infty$

**B.** $(x+2)^2$ is positive on both sides and $x - 1$ is near $-3$, so the quotient is negative on both sides. A: treated the even multiplicity as odd, so the sides differ. C: ignored the negative factor $x - 1$. D: odd-multiplicity pattern with the signs reversed.

**6** A rational function has vertical asymptotes at $x = -2$ and $x = 5$, a zero at $x = 1$, and no other zeros. Which could be its equation?

**A.** $r(x) = \dfrac{x+1}{(x-2)(x+5)}$

**B.** $r(x) = \dfrac{(x+2)(x-5)}{x-1}$

**C.** $r(x) = \dfrac{x-1}{(x+2)(x-5)}$

**D.** $r(x) = \dfrac{x-1}{(x-2)(x+5)}$

**C.** Factors $x+2$ and $x-5$ in the denominator give asymptotes at $-2$ and $5$; the factor $x-1$ in the numerator gives the zero at $1$. A: sign errors on every factor. B: swapped the numerator and the denominator. D: sign errors on the denominator factors only.

**7** The cost, in thousands of dollars, of removing $p$ percent of a pollutant is $C(p) = \dfrac{50p}{100-p}$ for $0 \le p < 100$. Which is the best interpretation of $\lim\limits_{p \to 100^-} C(p) = \infty$?

**A.** Removing $100\%$ of the pollutant costs exactly $100$ thousand dollars

**B.** The cost never exceeds $50$ thousand dollars

**C.** The cost decreases toward $0$ as the percentage removed approaches $100$

**D.** As the percentage removed gets close to $100$, the cost increases without bound

**D.** The denominator $100 - p$ is a tiny positive number and the numerator is near $5000$, so $C$ increases without bound. A: read $100$ as the value of the function. B: used the coefficient $50$ as a maximum. C: thought a small denominator gives a small output.

**8** Which gives all vertical asymptotes of $r(x) = \dfrac{(x-2)^2(x+1)}{(x-2)^3(x+4)}$?

**A.** $x = -4$ and $x = 2$

**B.** $x = -4$ only

**C.** $x = -4$, $x = -1$ and $x = 2$

**D.** $x = 4$ and $x = -2$

**A.** The zero $2$ has multiplicity $3$ in the denominator and $2$ in the numerator; $3 > 2$, so one factor $x-2$ remains: an asymptote. $x+4$ gives $x = -4$. B: cancelled the whole factor and called $x = 2$ a hole. C: also counted the numerator zero $x = -1$. D: sign errors on both factors.

AP PRECALCULUS · TOPIC 1.9 · SECTION 2

# Rational Functions and Vertical Asymptotes

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

**Question 9** Calculator allowed · 6 points

For a thin lens with focal length $10$ cm, an object placed $u$ cm from the lens ($u > 10$) forms an image $v$ cm from the lens, where $v(u) = \dfrac{10u}{u - 10}$.

**(a)** Find the equation of the vertical asymptote of the graph of $v$ and justify your answer. DOK3 2 pt

$u - 10 = 0$ at $u = 10$ and the numerator is $100 \neq 0$ there, so the vertical asymptote is $u = 10$.

**(b)** Write a limit statement describing the behavior of $v(u)$ as $u$ approaches $10$ from the right, and interpret it in context. DOK3 2 pt

$\lim\limits_{u \to 10^+} v(u) = \infty$. As the object gets closer to the focal point, the image distance increases without bound.

**(c)** Find the object distance $u$ for which the image distance is $25$ cm. Give your answer to three decimal places. DOK3 2 pt

$10u = 25(u - 10) \Rightarrow 15u = 250 \Rightarrow u = \dfrac{50}{3} \approx 16.667$ cm.

**Question 10** No calculator · 6 points

Let $r(x) = \dfrac{x^2 - 2x - 3}{x^2 - 2x}$.

**(a)** Factor the numerator and the denominator of $r$. Give the zeros and the equations of all vertical asymptotes. DOK3 2 pt

$r(x) = \dfrac{(x - 3)(x + 1)}{x(x - 2)}$. Zeros: $x = 3$ and $x = -1$. No factor cancels, so the vertical asymptotes are $x = 0$ and $x = 2$.

**(b)** Find $\lim\limits_{x \to 2^-} r(x)$ and $\lim\limits_{x \to 2^+} r(x)$. Show the sign of each factor. DOK3 2 pt

Near $x = 2$ the numerator is near $-3$ and $x$ is near $2 > 0$. From the left $x - 2 < 0$, so the quotient is $\dfrac{-3}{0^-}$: $\lim\limits_{x \to 2^-} r(x) = \infty$. From the right $x - 2 > 0$: $\lim\limits_{x \to 2^+} r(x) = -\infty$.

**(c)** Write an equation for a rational function $q$ with vertical asymptotes at $x = 0$ and $x = 2$ whose graph goes to the same infinity on both sides of $x = 0$. Justify your answer. DOK3 2 pt

For example $q(x) = \dfrac{1}{x^2(x - 2)}$. The factor $x^2$ has even multiplicity and is positive on both sides of $0$, and $x - 2$ is near $-2$, so $q \to -\infty$ from both sides of $x = 0$.

**Answer key (Section 1): 1A 2B 3C 4D 5B 6C 7D 8A**